INCH-POUND
MIL-R-24750
30 August 1990

# MILITARY SPECIFICATION

## ROPE, FIBROUS, SINGLE-BRAIDED 12 STRAND POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers 12 strand single braided polyester rope for general purpose use. The rope shall be of one type in the circumference specified (see 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| UU-T-81 | - Tags, Shipping and Stock. |
| PPP-F-320 | - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes. |

MILITARY

| | |
|---|---|
| MIL-C-3131 | - Cordage, Packaging of. |
| MIL-L-17331 | - Lubricating Oil, Steam Turbine and Gear, Moderate Service. |
| MIL-L-19140 | - Lumber and Plywood, Fire-Retardant Treated. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020
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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 885 - Methods of Testing Tire Cords, Tire Cord Fabrics, and Industrial Filament Yarns Made From Man-Made Organic-Base Fibers.
D 1141 - Standard Specification for Substitute Ocean Water. (DoD adopted)
D 1577 - Test Methods for Linear Density of Textile Fibers.
D 2257 - Standard Test Method for Extractable Matter in Textiles. (DoD adopted)
D 2258 - Standard Practice for Sampling Yarn for Testing.
D 4268 - Methods of Testing Fiber Ropes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Government, or supplier purchases. The requirements specified in 3.10 and 3.11 apply only to single braided polyester ropes purchased directly by the Government. Other requirements apply to single braided polyester ropes purchased as a component for an end item by a contractor and to single braided polyester ropes purchased directly by the Government. Splicing tools shall be furnished to the acquisition activity when required (see 6.2).



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3.3 Materials. The rope shall be fabricated from bright, white, virgin, continuous filament, heat and light resistant polyester fiber. Polyester fiber is a long chain synthetic polymer composed of at least 85 percent by weight of an ester of a substituted aromatic carboxylic acid, including but not restricted to substituted terephthalate units and para substituted hydroxy-benzoate units. The average fiber linear density per filament shall be not less than 4.0 denier per filament or more than 18 denier per filament. The fiber tenacity shall not be less than 8.0 grams per denier. The linear density per filament and the tenacity of the fiber shall be determined in accordance with ASTM D 885, ASTM D 1577, and ASTM D 2258.

3.4 Construction. The rope shall be a single braided construction consisting of 12 strands in a twill pattern where one strand of one direction of rotation about the axis of the rope passes over two strands of the opposite direction and then passes under the next two strands of the opposite direction. One half of the strands shall be "Z" twist while the remaining half shall be "S" twist. Heat setting of the yarns, strands, or finished rope shall not be acceptable.

3.4.1 Strands. Rope strands shall be one size and shall have sufficient twist such that the yarns on the outer surface of the strand shall be essentially parallel to the axis of the rope.

3.4.2 Texture. The texture of the finished rope shall be determined by the multipliers contained in table I. For sizes 3/8 inch diameter through 1 inch diameter the multiplier shall ensure spliceability using a tuck splice or a hollow braid splice. For sizes above 1 inch diameter the multiplier shall ensure spliceability using a tuck splice. The multiplier shall also ensure that the rope texture will not be so loose as to cause undue snagging of the strands. The use of multipliers overcomes the complexity of listing a range of pick counts for every size. The multiplier is calculated as follows:

$$\text{Multiplier} = \text{Pick Count} \times \text{Circumference}$$

Pick count and circumference of the rope shall be measured at the appropriate load P (table II) for the nominal rope size and shall be used in all calculations.

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TABLE I. Rope structural requirements.

| | Type of braid | Twill |
|---|---|---|
| | Number of strands | 12 |
| | Yarns per strand (minimum) | 1 |
| | Yarns per strand (maximum) | 2 |
| | 3/8 through 1 inch diameter | Above 1 inch diameter |
| Multiplier (minimum) | 3.8 | 4.0 |
| Multiplier (maximum) | 4.2 | 5.2 |

TABLE II. Physical properties.

| Nominal circumference at load P (inches) | Nominal diameter (inches) | Load P (pounds) | Linear density plus or minus 5 percent (pounds per 100 feet) | Minimum breaking strength (pounds) |
|---|---|---|---|---|
| 1-1/8 | 3/8 | 28 | 4.2 | 4240 |
| 1-5/16 | 7/16 | 38 | 5.7 | 5680 |
| 1-1/2 | 1/2 | 50 | 7.7 | 7440 |
| 1-3/4 | 9/16 | 63 | 9.8 | 9280 |
| 2 | 5/8 | 78 | 11.5 | 11520 |
| 2-1/4 | 3/4 | 113 | 16.6 | 16640 |
| 2-3/4 | 7/8 | 153 | 22.6 | 19440 |
| 3 | 1 | 200 | 30 | 25600 |
| 3-1/2 | 1-1/8 | 253 | 38 | 32800 |
| 3-3/4 | 1-1/4 | 313 | 44 | 37600 |
| 4 | 1-5/16 | 345 | 50 | 43200 |
| 4-1/2 | 1-1/2 | 450 | 63 | 54400 |
| 5 | 1-5/8 | 528 | 78 | 67200 |
| 5-1/2 | 1-3/4 | 613 | 94 | 80800 |
| 6 | 2 | 800 | 112 | 96000 |
| 6-1/2 | 2-1/8 | 903 | 132 | 113600 |
| 7 | 2-1/4 | 1013 | 153 | 131200 |
| 7-1/2 | 2-1/2 | 1250 | 176 | 151200 |
| 8 | 2-5/8 | 1378 | 200 | 172000 |
| 9 | 3 | 1800 | 253 | 215200 |
| 10 | 3-1/4 | 2113 | 312 | 264800 |
| 11 | 3-1/2 | 2450 | 378 | 319200 |
| 12 | 4 | 3200 | 449 | 376800 |
| 13 | 4-1/4 | 3613 | 527 | 440800 |
| 14 | 4-1/2 | 4050 | 612 | 508800 |
| 15 | 5 | 5000 | 702 | 581600 |



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3.5 Physical requirements. The finished rope shall conform to the physical properties specified in table II when tested as specified in 4.5 and 4.6. The circumference in accordance with table II shall be as specified (see 6.2).

3.5.1 Elongation. The elongation of the rope shall not exceed 30 percent at the minimum breaking strength when determined as specified in 4.6.5. The load elongation curve, drawn autographically, shall not exhibit evidence of sudden changes in load applications greater than 5 percent of the load measured at the instant of change. Changes due to splice slippage shall not be considered in this determination.

3.6 Finish. No extraneous material shall be added for purposes of weighting the rope. The extractable matter of the finished rope shall not exceed 5 percent when tested as specified in 4.6.6.

3.7 Moisture content. The moisture content of the rope shall not exceed 4 percent when tested as specified in 4.6.8.

3.8 Identification marker. The manufacturer shall identify his product by inserting a Kraft paper or water resistant marker inside one strand in all ropes larger than 1-1/8 inches circumference. The marker shall be completely enveloped by the cover yarns in the strand in which enclosed. Unless otherwise specified (see 6.2), the manufacturer's name, the year of the manufacture, and type of fiber (POLYESTER) shall be printed on the marker in bold, easy to read type. Italic or script type shall not be used. The printing shall not be effected by exposure to salt water or mineral oil, when tested as specified in 4.6.7.

3.9 Identification tag. In addition to the requirements specified in 3.8, each package unit shall have a ticket (identification tag) attached for identification purposes. The ticket shall conform to the requirements of type B, class 1, size 4 or 5, 15 CSU grade of UU-T-81. The ticket shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information:

(a) Stock number.
(b) Nomenclature.
(c) Specification number.
(d) Length.
(e) Contract number and date.
(f) Date of manufacture and year.
(g) Contractor's name.

3.10 Put-up. Unless otherwise specified (see 6.2), the rope shall be furnished without knots or splices on nonreturnable reels not larger that 7 feet in diameter in lengths specified in table III. Multiple lengths shall be permitted but no piece shall be less than 600 feet when measured in the relaxed condition. The rope shall be ordered by weight as specified (see 6.2). Delivered weight per reel shall be within plus 10 percent, minus 0 percent of the weight specified in table III. The weight for lengths other than standard shall be determined in accordance with table III with the proper length ratio applied.



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TABLE III. Put-up.

| Circumference (inches) | Diameter (inches) | Standard length (feet) | Net weight per standard length (pounds) |
|---|---|---|---|
| 1-1/8 | 3/8 | 600 | 26.5 |
| 1-5/16 | 7/16 | 600 | 35.9 |
| 1-1/2 | 1/2 | 600 | 46.6 |
| 1-3/4 | 9/16 | 600 | 58.6 |
| 2 | 5/8 | 600 | 72.5 |
| 2-1/4 | 3/4 | 600 | 104.6 |
| 2-3/4 | 7/8 | 600 | 142.4 |
| 3 | 1 | 600 | 189 |
| 3-1/2 | 1-1/8 | 600 | 239 |
| 3-3/4 | 1-1/4 | 600 | 277 |
| 4 | 1-5/16 | 600 | 315 |
| 4-1/2 | 1-1/2 | 600 | 397 |
| 5 | 1-5/8 | 600 | 491 |
| 5-1/2 | 1-3/4 | 600 | 592 |
| 6 | 2 | 600 | 706 |
| 6-1/2 | 2-1/8 | 600 | 832 |
| 7 | 2-1/4 | 600 | 964 |
| 7-1/2 | 2-1/2 | 600 | 1109 |
| 8 | 2-5/8 | 600 | 1260 |
| 9 | 3 | 600 | 1594 |
| 10 | 3-1/4 | 600 | 1966 |
| 11 | 3-1/2 | 600 | 2381 |
| 12 | 4 | 600 | 2829 |
| 13 | 4-1/4 | 600 | 3320 |
| 14 | 4-1/2 | 600 | 3856 |
| 15 | 5 | 600 | 4423 |

3.10.1 Braider splices. Braider splices shall be the overlapping continuation of a single interrupted strand with another identical strand which follows the identical path in the rope. Although it is desirable that no braider splices be present in any size and length of rope, some methods of manufacture impose limitations. To compensate for these limitations, the following shall be acceptable:

3.10.1.1 Standard length. To allow for a braider malfunction, one braider splice shall be permitted for orders specifying standard length or less as listed in table III.

3.10.1.2 Continuous lengths greater than standard lengths. To compensate for machinery limitations when continuous lengths are greater than standard lengths and up to 1200 feet maximum, one braider splice shall be permitted in each major strand for rope sizes 1-1/8 inch through 12 inch, and two braider splices shall be permitted in each major strand for rope sizes 13 inch through 15 inch. In producing the splices the distance of the overlapping shall be equivalent to eight times the circumference in inches but not less than 24 inches for ropes whose sizes are 3 inches in circumference and less.



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3.11 Workmanship. The ends of all rope shall be cut off squarely and be securely whipped, taped, or heat sealed. The reels shall be wound so that each turn and layer is free from entanglement. The rope shall conform to the quality and grade of product established by the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations specified in 4.5.1 and 4.5.2 and tests specified in 4.6.

4.3.1 First article sample. The first article sample for ropes shall be a continuous finished length of 70 feet.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations specified in 4.5.1 and 4.5.2 and the tests specified in 4.4.1 and 4.6 (see 6.3). Unless otherwise specified herein, sampling for inspection shall be performed in accordance with tables IV and V.



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TABLE IV. Sampling for visual examination and testing of the end item.

| Lot size | Sample size |
|---|---|
| 2 - 15 | 2 |
| 16 - 25 | 3 |
| 26 - 90 | 5 |
| 91 - 150 | 8 |
| 151 - 280 | 13 |
| 281 - 500 | 20 |
| 501 - 1200 | 32 |
| 1201 - 3200 | 50 |
| 3201 - 10000 | 80 |

TABLE V. Sampling for examination for length and winding.

| Lot size | Sample size |
|---|---|
| 1 - 10 | All |
| 11 - 500 | 10 |
| 501 - 3200 | 13 |
| 3201 - 10000 | 20 |

4.4.1 Material and component inspection. Determination shall be made for all characteristics specified in table VI, except material. A determination by the fiber supplier for material and by the rope contractor attesting to the number of braider splices shall be considered acceptable for these characteristics (see 6.3 and appendix). Results shall be determined as pass or fail.

TABLE VI. Component testing.

| Component | Requirement | Test method | Number of determinations per individual sample unit |
|---|---|---|---|
| Fiber: | | | |
| Type | 3.3 | 1/ | 1 |
| Linear density | 3.3 | ASTM D 1577 | 1 |
| Tenacity | 3.3 | ASTM D 885 | 1 |
| Rope braid | | | |
| Yarn size | 3.4.1 | Visual | 1 |
| Yarns/strand | Table I | Visual | 1 |
| Number of strands | Table I | Visual | 1 |
| Braid type | Table I | Visual | 1 |
| Multiplier | Table I | 3.4.2 | 3 |
| Heat setting | 3.4 | 1/ | 1 |
| Braider splices | 3.10.1 | 1/ | 1 |

1/ See 4.4.1 and appendix.



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4.5 Examinations.

4.5.1 End item for visual defects. The unit of product for this examination shall be one reel. Ten percent of the gross length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels each. The sample units will be randomly selected, and the sample size shall be as specified in table IV. Any sample unit having one or more defects specified in table VII shall be rejected. If any defects are noted in the original sample units, additional units shall be randomly selected and if any defects specified in table VII are noted the entire lot shall be rejected.

TABLE VII. Visual defects.

| Defects |
|---|
| Appearance and workmanship |
| Cut, any. |
| Chafed or damaged. |
| Kinks or uneven braiding resulting in braid discontinuity, or soft spots. |
| Broken, loose, or projecting ends in the rope or strand. |
| Excessive loose fiber ends on surface, or in gaps between strands. |
| Ends not cut off squarely. |
| Ends not securely whipped, taped, or heat sealed. |
| Identification marker 1/ |
| Omitted, incorrect, illegible. |
| Italic or script type used, not as specified. |
| Not completely covered in strand. |
| Identification ticket |
| Omitted, incorrect, illegible. |
| Insecurely attached. |
| Hand written entries. |
| Cleanliness |
| Spot or stain clearly visible. 2/ |

1/ Identification marker for braids larger than 1-1/8 inch circumference.
2/ At normal inspection distance (approximately 3 feet).

4.5.2 Length and winding. The sample unit for this examination shall be one reel. The sample size shall be as specified in table V. The lot size shall be the number of units in the inspected lot. Defects shall be as specified in 4.5.2.1 and 4.5.2.2.

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4.5.2.1 Defects in length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

(a) Length of unit less than or more than length specified (including permitted tolerances).
(b) Length of unit less than marked on ticket.
(c) Reels not in continuous length.
(d) Any piece on reels less than 600 feet in length for ropes with a circumference of 3/4 inch up to 15 inches.

4.5.2.2 Defects in winding. Defects with regard to winding shall be considered to exist if the following are determined during inspection:

(a) Improperly or not firmly wound resulting in slippage during unwinding or otherwise affecting free unhampered unwinding of the rope.
(b) Rope ends knotted or spliced to make a continuous length, except as noted in 3.10.1.1 and 3.10.1.2.

4.6 Testing of the end item. Testing of the end item shall be as specified in table VIII and 4.6.2 through 4.6.8. The physical and chemical values specified herein apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size, shall be in accordance with table IV and shall be 0 percent defective. The lot size shall be expressed in units of reels. The sample unit for test purposes shall be 70 feet of rope.

TABLE VIII. End item testing.

| Rope property | Requirement | Test method | Number of determinations per individual sample unit |
|---|---|---|---|
| Circumference | 3.5 | 4.6.2.1 | 3 |
| Linear density | 3.5 | 4.6.3 | 1 |
| Breaking strength | 3.5 | 4.6.4 | 2 (up to 9-inch circ.)<br>1 (10-inch circ. & above) |
| Elongation | 3.5.1 | 4.6.5 | 2 |
| Extractable matter | 3.6 | 4.6.6 | 2 |
| Identification marker | 3.8 | 4.6.7 | 2 |
| Moisture content | 3.7 | 4.6.8 | 2 |

4.6.1 Specimen preparation. The designated length and the number of test specimens shall be removed from the selected test reels in accordance with ASTM D 4268.



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4.6.2 Circumference and picks per inch.

4.6.2.1 Circumference. The circumference shall be measured in accordance with ASTM D 4268. Results to be reported shall be the average of three determinations to the nearest 1/16 inch.

4.6.2.2 Pick count. While under load P ten complete picks shall be counted and marked off. The distance between the marks shall be measured to the nearest 1/16 inch. This procedure shall be repeated at least three times in different positions (not less than 1 foot of rope apart). The multiplier value shall be calculated for the respective sizes by dividing 10 by the observed dimension and multiplying the result by the circumference.

4.6.3 Linear density.

4.6.3.1 Density determination. The linear density shall be determined in accordance with ASTM D 4268. Tests to determine compliance with the requirements (including quantity for delivery) specified herein, may be conducted under prevailing atmospheric conditions. In case of dispute, the tests shall be conducted on material which has reached equilibrium under standard conditions specified in ASTM D 4268.

4.6.3.2 Accuracy. Linear density results shall be determined as follows:

| Circumference (inches) | Degree of accuracy (pounds per 100 feet) |
|---|---|
| Less than 1-3/4 | Nearest 0.01 |
| 2 to 6 | Nearest 0.10 |
| 6-1/2 to 12 | Nearest 1.0 |
| More than 12 | Nearest 10.0 |

4.6.4 Breaking strength. Breaking strength determinations shall be conducted in accordance with ASTM D 4268. Results reported shall be the average to an accuracy of 1 percent. The load applied need not cause a failure provided that all individual test values exceeds the specified minimum breaking strength.

4.6.5 Elongation. The elongation at the minimum breaking strength shall be determined in accordance with the initial single loading procedure of ASTM D 4268. The results reported shall be the average of the determination to the nearest 0.1 percent.

4.6.6 Extractable matter. The extractable content shall be determined in accordance with ASTM D 2257, using a Soxhlet apparatus in the procedure. Results reported shall be the average of two determinations to the nearest 0.1 percent.

4.6.7 Identification marker. Three lengths of the identification marker shall be used to determine the fastness of the printed matter of the identification marker to saltwater and mineral oil. Each length shall be approximately 1-1/2 feet. One length shall be retained as a control, one length shall be immersed for 2 hours in synthetic seawater conforming to ASTM D 1141, stock solution number 1, 2, or 3, and one length shall be soaked for 2 hours in mineral oil conforming to symbol 2190-TEP of MIL-L-17331. Following removal from the

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respective environments, the two exposed specimens shall be visually compared with the control specimen. The fastness of the printed matter shall be considered satisfactory when no perceptible change in color or legibility is observed.

4.6.8 Moisture content. The moisture content shall be determined in accordance with method 2600 of FED-STD-191. Results reported shall be the average of two determinations to the nearest 0.1 percent.

4.7 Inspection of packaging. Sample packages and the inspection of the preservation packing, and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition for the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container, reel (spools), and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of the interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class domestic/fire retardant or class weather resistant/fire retardant materials requirements as specified (see 6.2) of PPP-F-320 and amendments thereto.

5.2 Packaging requirements. The packaging (preservation, packing and marking) requirements shall be in accordance with MIL-C-3131 for the level (A, C or commercial) of preservation except as specified herein for 13 inch through 16 inch circumference rope, level (A, B, C, or commercial) of packing, and marking including packaging acquisitioning options therein as specified (see 6.2). For level A preservation of 13 inch through 16 inch circumference, rope, reels shall be constructed in accordance with the wood and plywood requirements specified in MIL-C-3131 and the following:



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(a) Wood.
(1) The thickness of the reel head boards shall be minimum 1-inch, two or three ply.
(2) The barrel shall consist of minimum 1-1/2 inch thick wood staves mortised into the reel heads.
(3) The reel shall be assembled with a minimum of five, 1/2 or 3/4 inch diameter steel bolts having cup washers on each end.

(b) Plywood.
(1) 1-1/4 inch, two sheets of 3/4-inch, thick for net weights 700 through 1675 pounds; in addition, minimum of five, 1/2 inch nuts and bolts shall be used for the flange and barrel assembly.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rope covered by this specification is intended for general use where high strength, low stretch, and good abrasion resistance are required.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Circumference size required (see 1.1 and 3.5).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article is required (see 3.1).
(e) When splicing tools are required (see 3.2).
(f) When information on the identification marker is other than specified (see 3.8).
(g) When multiple lengths per unit are not allowed (see 3.10).
(h) Put-up if other than specified (see 3.10).
(i) When continuous lengths, other than standard, are required (see 3.10.).
(j) Levels of preservation, packing and marking required (see 5.2).
(k) When fire retardant requirements are required (see 5.1.1).
(l) That the purchaser will accept at original weight any unit which has been shortened or cut for test specimens if in complete compliance with this specification. Reel shall be marked as to actual length.
(m) That single braid polyester will be purchased on a price-per-pound basis net weight.



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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 4.4 | DI-T-2072 | Reports, test | ---- |
| 4.4.1 and appendix | DI-MISC-80678 | Certification/data report | ---- |

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article inspection. When a first article inspection is required the items should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in the acquisition test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 5-percent moisture basis.

6.6 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are required by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Cord
Pick-count
Splices

Custodians:
Army - GL
Air Force - 99
Navy - SH

Review activities:
Air Force - 82

Preparing activity:
Navy - SH
(Project 4020-0312)



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APPENDIX

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included on technical reports when required by the contract or purchase order. This appendix is mandatory only when data item description DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION CONTENT

30.1 Certificate of compliance. A certificate of compliance shall contain the following information (see 4.1.1):

(a) Conformance of material characteristics to the requirements specified herein (see 3.3).
(b) Conformance to the requirement for heat setting of yarns or finished rope (see 3.3).
(c) Conformance to the requirement for the mixing of fiber types, fiber grades, or fibers of different manufacturers within the rope (see 3.4).
(d) Number of braider splices used (see 3.10.1.1).



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-R-24750 | 30 August 1990 |

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| Technical point of contact (TPOC):<br>Jack Hall (SEA 56Y23)<br>PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | TPOC: 703-602-1844 | 8-332-1844 |
| c. ADDRESS (Include Zip Code)<br>Commander, Naval Sea Systems Command<br>Department of the Navy (SEA 55Z3)<br>Washington, DC 20362-5101 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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